SEQUENCE OF OPERATION

FAN COIL UNIT

SETPOINT CONTROL

THE BAS SHALL EDIT THE ZONE SPACE TEMPERATURE SETPOINT OF EACH FAN COIL UNIT. THE ZONE TEMPERATURE SETPOINT SHALL BE OPERATOR ADJUSTABLE. INDIVIDUAL ZONE SETPOINT AND CONTROL LOGIC SHALL RESIDE AT THE ZONE LEVEL, AND NOT BE DEPENDENT UPON THE BAS FOR CONTROL. IN THE EVENT OF COMMUNICATION LOSS, THE BOX WILL CONTINUE TO CONTROL TO CURRENT SETPOINTS.

OPERATING MODE

THE BAS SHALL PLACE THE FAN COIL UNIT IN EITHER THE OCCUPIED OR UNOCCUPIED MODE BASED ON AN OPERATOR ADJUSTABLE TIME SCHEDULE. SEPARATE HEATING AND COOLING SETPOINTS SHALL BE ENTERABLE FOR EACH MODE THROUGH THE BAS.

OCCUPIED - HEATING

THE UNIT FAN SHALL START ON A SIGNAL FROM THE BAS THROUGH THE START TIME OPTIMIZATION PROGRAM. ON A FALL IN TEMPERATURE BELOW THE HEATING SET POINT THE HOT WATER COIL CONTROL VALVE SHALL MODULATE OPEN TO MAINTAIN THE ROOM TEMPERATURE SET POINT. ON A RISE IN TEMPERATURE, THE REVERSE SHALL OCCUR.

OCCUPIED - COOLING

THE UNIT FAN SHALL START ON A SIGNAL FROM THE BAS THROUGH THE START TIME OPTIMIZATION PROGRAM. ON A RISE IN TEMPERATURE ABOVE THE COOLING SET POINT THE CHILLED WATER COIL CONTROL VALVE SHALL MODULATE OPEN TO MAINTAIN THE ROOM TEMPERATURE SET POINT. ON A FALL IN TEMPERATURE, THE REVERSE SHALL OCCUR.

UNOCCUPIED - HEATING

THE BAS SHALL CYCLE THE UNIT FAN TO MAINTAIN THE UNOCCUPIED HEATING SET POINT. THE HOT WATER COIL CONTROL VALVE SHALL BE FULLY OPEN TO THE COIL. EACH FAN COIL UNIT SHALL MAINTAIN THE NIGHT SETBACK TEMPERATURE SET POINT FOR ITS AREA SERVED.

UNOCCUPIED - COOLING

THE BAS SHALL CYCLE THE UNIT FAN TO MAINTAIN THE UNOCCUPIED COOLING SET POINT. THE CHILLED WATER CONTROL VALVE SHALL BE FULLY OPEN TO THE COIL. EACH FAN COIL UNIT SHALL MAINTAIN THE NIGHT SETBACK TEMPERATURE SET POINT FOR ITS AREA SERVED.

NIGHT SETBACK OVERRIDE

A NIGHT SETBACK OVERRIDE PUSHBUTTON ON THE SPACE TEMPERATURE SENSOR SHALL OVERRIDE THE UNOCCUPIED MODE AND RETURN THE UNIT TO THE OCCUPIED MODE OF OPERATION. THIS SHALL ACTIVATE THE ASSOCIATED SUPPLY AIR FAN AND THE REQUIRED HEATING/COOLING EQUIPMENT. OVERRIDE SHALL BE ADJUSTABLE FROM 1 TO 6 HOURS. OVERRIDE SHALL RETURN ENOUGH UNITS TO THE OCCUPIED MODE TO MAINTAIN LOAD ON THE CHILLER.

SENSOR OVERRIDE CAPABILITY SHALL BE ACTIVATED OR DEACTIVATED AT THE MAIN PANEL AS REQUIRED BY THE OWNER.

WARM-UP AND COOLDOWN

DURING WARM-UP AND COOLDOWN OPERATION AS DETERMINED BY THE START TIME OPTIMIZATION PROGRAM, THE UNIT FAN SHALL RUN CONTINUOUSLY AND THE RESPECTIVE COIL CONTROL VALVE SHALL MODULATE FULLY OPEN IN ORDER TO REACH THE OCCUPIED SET POINT PRIOR TO THE SCHEDULED OCCUPANCY. THE RESPECTIVE SUPPLY AIR FAN SHALL REMAIN OFF DURING WARM-UP AND COOLDOWN.

CHILLERS

ON A CALL FOR COOLING, THE CHILLED WATER PUMP SHALL BE STARTED AND THE CHILLER SHALL BE ENABLED. A CALL FOR COOLING IS DETERMINED BY THE BUILDING HEATING/COOLING MODE SELECTION. ON A RISE IN CHILLED WATER SUPPLY TEMPERATURE, THE CHILLER SHALL START. THE CHILLER SHALL MODULATE BASED ON CHILLED WATER SUPPLY TEMPERATURE. CHILLED WATER SUPPLY TEMPERATURE SHALL BE MAINTAINED AT 45 DEGREES F.

CHILLED WATER PUMPS

THE LEAD CHILLED WATER PUMP SHALL BE STARTED PRIOR TO STARTING THE CHILLER.

THE DDC CONTROLLER SHALL ALTERNATE PUMPS WEEKLY BY SELECTING THE PUMP WITH THE LEAST TOTAL ACCUMULATED RUN TIME AS CALCULATED AND STORED AT THE DDC CONTROLLER. IF EITHER SELECTED LEAD PUMP'S PROOF SWITCH DOES NOT DETECT FLOW WITHIN AN ALLOWABLE TIME FRAME, THE DDC CONTROLLER SHALL DE-ENERGIZE THE FAILED PUMP, START THE STANDBY PUMP, AND AN ALARM SHALL BE SENT TO THE DDC CONTROLLER. A PROGRAMMED ANTI-RECYCLE TIMER SHALL PREVENT PUMP SHORT CYCLING.

THE DDC CONTROLLER SHALL AUTOMATICALLY ATTEMPT A RESTART OF THE CIRCULATING PUMPS ON A PERIODIC BASIS TO ACCOMMODATE SERVICE SHUTDOWNS AND EXTENDED POWER OUTAGES.

CONVERTER

ON A FALL IN TEMPERATURE AT THE HOT WATER SUPPLY SENSOR BELOW SET POINT, THE STEAM CONVERTER CONTROL VALVE SHALL START TO MODULATE TOWARDS THE FULL OPEN POSITION. ON A CONTINUED FALL IN TEMPERATURE BELOW SET POINT, STEAM CONVERTER CONTROL VALVE SHALL CONTINUE TO MODULATE OPEN. ON A RISE IN HOT WATER TEMPERATURE THE REVERSE SHALL OCCUR.

THE PRIMARY HOT WATER SUPPLY TEMPERATURE SET POINT SHALL BE RESET ACCORDING TO A PREDETERMINED SCHEDULE BASED ON OUTSIDE AIR TEMPERATURE. THE DDC CONTROLLER SHALL MODULATE THE STEAM CONVERTER CONTROL VALVE TO MAINTAIN THE WATER TEMPERATURE SET POINT.

HOT WATER PUMPS

THE LEAD HOT WATER PUMP SHALL BE STARTED PRIOR TO ENABLING THE CONVERTER STEAM CONTROL VALVE.

THE DDC CONTROLLER SHALL ALTERNATE PUMPS WEEKLY BY SELECTING THE PUMP WITH THE LEAST TOTAL ACCUMULATED RUN TIME AS CALCULATED AND STORED AT THE DDC CONTROLLER. IF EITHER SELECTED LEAD PUMP'S PROOF SWITCH DOES NOT DETECT FLOW WITHIN AN ALLOWABLE TIME FRAME, THE DDC CONTROLLER SHALL DE-ENERGIZE THE FAILED PUMP, START THE STANDBY PUMP, AND AN ALARM SHALL BE SENT TO THE DDC CONTROLLER. A PROGRAMMED ANTI-RECYCLE TIMER SHALL PREVENT PUMP SHORT CYCLING.

THE DDC CONTROLLER SHALL AUTOMATICALLY ATTEMPT A RESTART OF THE CIRCULATING PUMPS ON A PERIODIC BASIS TO ACCOMMODATE SERVICE SHUTDOWNS AND EXTENDED POWER OUTAGES.

HEATING-COOLING CHANGEOVER

HEATING

WHEN THE OUTSIDE AIR TEMPERATURE FALLS BELOW 62 DEGREES F (ADJUSTABLE), THE SYSTEM SHALL BE INDEXED TO THE HEATING MODE. WHEN INDEXED TO THE HEATING MODE, THE CHILLER AND THE CHILLED WATER PUMP SHALL BE DISABLED. THE HOT WATER PUMP SHALL BE STARTED AND THE STEAM CONVERTER CONTROL VALVE SHALL BE UNDER CONTROL.

COOLING

WHEN THE OUTSIDE AIR TEMPERATURE RISES ABOVE 65 DEGREES F (ADJUSTABLE), THE SYSTEM SHALL BE INDEXED TO THE COOLING MODE. WHEN INDEXED TO THE COOLING MODE, THE STEAM CONVERTER VALVE SHALL BE FULLY CLOSED AND THE HOT WATER PUMP SHALL BE DISABLED. WHEN THE CHILLED WATER RETURN TEMPERATURE FALLS BELOW 85 DEGREES F, THE LEAD CHILLED WATER PUMP AND CHILLER SHALL BE STARTED AND OPERATE AS DESCRIBED ABOVE.

EXHAUST AND SUPPLY FANS

THE INDICATED EXHAUST AND SUPPLY AIR FANS SHALL BE SCHEDULED BY THE BUILDING OCCUPANCY SCHEDULE.

THE INDICATED EXHAUST FANS SHALL BE STARTED BY A WALL MOUNTED THERMOSTAT SUCH THAT WHEN THE SPACE TEMPERATURE RISES ABOVE THE SET POINT, THE EXHAUST FAN SHALL BE ENERGIZED.

ON EF-5 ONLY, RESPECTIVE INTAKE DAMPER AND DAMPER AT FAN SHALL OPEN WHEN FAN IS ON. OTHERWISE, DAMPERS SHALL BE CLOSED.

UNIT HEATERS

UNIT HEATERS SHALL BE CYCLED BY WALL MOUNTED THERMOSTATS.

HOT WATER COIL (AT EXISTING CREDIT UNION)

THE SPACE THERMOSTAT SHALL MODULATE THE HOT WATER COIL CONTROL VALVE TO MAINTAIN THE HEATING SETPOINT. ON A FALL IN SPACE TEMPERATURE THE HOT WATER CONTROL VALVE SHALL BE MODULATED OPEN TO THE COIL. ON A FALL IN SPACE TEMPERATURE THE REVERSE SHALL OCCUR.

M-12

PACE COLLABORATIVE	HUDSON + ASSOCS, ARCHITECTS 503 BOTETOURT STREET NORFOLK, VA. 23510	DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND NAVY PUBLIC WORKS CENTER NAVAL STATION — NORFOLK, VA
	JOB ORDER NO. 1323093	CINCLANT AREA — NORFOLK, VA.
	DES. NKC / DR. WGB / CHK. PDP	ADDITIONS & RENOVATIONS TO BUILDINGS NH-15, NH-16, & NH-17
	BR. MGR. J.H. SALLEY, JR.	
COMMONWEALTH OF VIRGINIA PATRICK D. PROPST No. 18267 10-14-94 9315 PROFESSIONAL ENGINEER	SUBMITTED BY: DATE:	
	EIC J. STARK	SEQUENCE OF OPERATION
	DIV. DIR. R.H. IVES, JR.	
	APPROVED: CAPT. DJ FELTES (SCUSN) DATE: 10/14/94 DR.	DR. D / CODE IDENT NO. 80091 / NAVFAC DRAWING NO. 4,287,836 / PWC DRAWING NO. 15060GB
	APPROVED: DATE: 10/26/94 FOR C.O. FOR COMMAND OFFICER, NAVFAC	CONSTR. CONTR. NO. N62470-94-B-2232
		SCALE: NONE / SPEC 05-94-2232 / SHEET 60 OF 78

DRAWING DATE